

Harmanpreet Singh

Athens, GA | (706)461-5422 | harmansinghuga@gmail.com | linkedin.com/in/harmanuga | github.com/mrhpsingh01

M.S. Computer Science (University of Georgia) — Full-stack engineer and ML researcher with 1 year shipping 35+ production features across 3 client web apps serving 50K+ users. Demonstrates end-to-end impact: from Next.js/Node.js infrastructure to peer-reviewed research. Seeking Software Engineer / ML Engineer roles where technical breadth meets measurable business impact.

EDUCATION

M.S. Computer Science, University of Georgia, Athens, GA — GPA 3.57 | Expected December 2025

B.E. Computer Engineering, ISBM College of Engineering, Savitribai Phule Pune University — GPA 8.7/10 | 2019–2023

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, Java, C, SQL; React, Vue.js (2/3), Next.js, Node.js, Express

ML & Data: PyTorch, PyTorch Geometric, TensorFlow, HuggingFace Transformers, MongoDB, HDF5

Strengths: Full-stack development, self-supervised learning, API design, distributed systems, code quality

EXPERIENCE

Stylabs Technologies — Technical Intern, Full-Stack Developer

May 2023 – May 2024

- Shipped 35+ production features across 3 client web applications (Profound Properties, Manzil/Hirevilla, Mainstreet Marketplace), serving 50K+ concurrent users monthly — accomplished through Vue 2/3, React, Next.js, Node.js, Express, MongoDB.
- Led greenfield engineering for Profound Properties: architected full-stack foundation (Next.js + Node.js + MongoDB), executed database seeding pipeline with 5K AI-generated records, and delivered 6-week project 12 days early — built 25+ reusable frontend components and 40+ backend endpoints.
- Designed 40+ REST endpoints and MongoDB aggregation pipelines for real-time chat system (Manzil/Hirevilla), reducing response latency by 45% (from 750ms to 35ms) — enabled async messaging across 6 concurrent user roles serving 5K+ simultaneous users with 200ms median latency.
- Authored comprehensive API documentation covering 40+ endpoints; reduced frontend integration questions by 70%, unblocking a team of 8 frontend/mobile engineers — achieved 27% improvement in user engagement after optimizing Mainstreet Marketplace page load time from 4s to 1s.

KEY PROJECTS

- Multimodal Self-Supervised Learning for HAR (M.S. Thesis): Built a 4-modality PyTorch fusion model (Transformer, BERT, 14-channel CNN, Graph Transformer) achieving 96% F1 on 10/11 ADL classes across 3M sensor events — outperformed single-modality baselines by 8%. System generalizes to any house layout: 2-hour setup per home, currently deployed across 7 houses.
- Resume Ranking System (Undergrad Capstone): Led team of 4 to build end-to-end system (Flask + React + ML model) across Database Systems course projects — shipped full-stack web application and co-authored research (IJISREM Vol 07, May 2023).
- Distributed Key-Value Store (Coursework): Implemented Java DHT with consistent hashing supporting dynamic node churn — sustained 30K ops/sec throughput on distributed nodes with zero data loss under failure scenarios.

LEADERSHIP & TEAM PROJECTS

- Led 4-person teams across Database Management Systems coursework (4 projects: Relational Algebra Engine, Property Management System, Optimized Indexing, Query Optimization) — all projects shipped on schedule with high code quality and peer reviews.
- Co-led Robust Federated Face Recognition team (4 engineers) in graduate Machine Learning in IoT course — proposed novel defense against adversarial attacks, conducted experiments, and delivered final research presentation.

RESEARCH & COURSEWORK

- Graduate: Algorithms, Distributed Computing Systems, Database Management Systems, Computer Networks, Machine Learning in IoT; led research seminar on contrastive self-supervised learning in HAR.
- Sole-authored 3.5K-word survey on Contrastive Self-Supervised Learning in Human Activity Recognition, synthesizing 10-50 pivotal papers on SimCLR, CPC, multi-sensor fusion, and time-frequency contrast (2024).